

Project Executable Files

Date	28 June 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	3 Marks

Project Executable Files

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	No
6	Deployed application URL (if hosted)	NA
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Credit Card Approval Prediction/

```

├── app.py
├── train.py
├── random_forest_model.pkl
├── clean_dataset.csv
├── requirements.txt
├── README.md
├── .gitignore
├── templates/
│   ├── index.html
│   └── result.html
├── static/
│   ├── style.css
│   └── images/
├── screenshots/
│   ├── Home_Page.png
│   ├── Approved_Result.png
│   └── Rejected_Result.png
├── test_cases/
│   └── Test_Results.pdf
├── documentation/
│   ├── Project_Report.pdf
│   └── Demo_Video_Link.txt

```

Step 3: Deployment / Access Details

Field	Details
Deployed URL	4037induja.pythonanywhere.com/
Login Credentials (Demo)	243ta04037@recw.ac.in 243TA04037
Platform / Hosting Provider	Local System (Flask)
Repository Link	https://github.com/243ta04037-induja/Credit-Card-Approval-Prediction/new/main
Demo Video Link	https://drive.google.com/file/d/1rd8mP0BBYL9UmlqOc_1YTfkfZ32oMFhl/view?usp=sharing

Step 4: Run Instructions

1. Install **Python 3.x** on your system.
2. Open the **Credit Card Approval Prediction** project folder in **Visual Studio Code**.
3. Install the required Python libraries using:

```
pip install -r requirements.txt
```

4. Run the Flask application using:

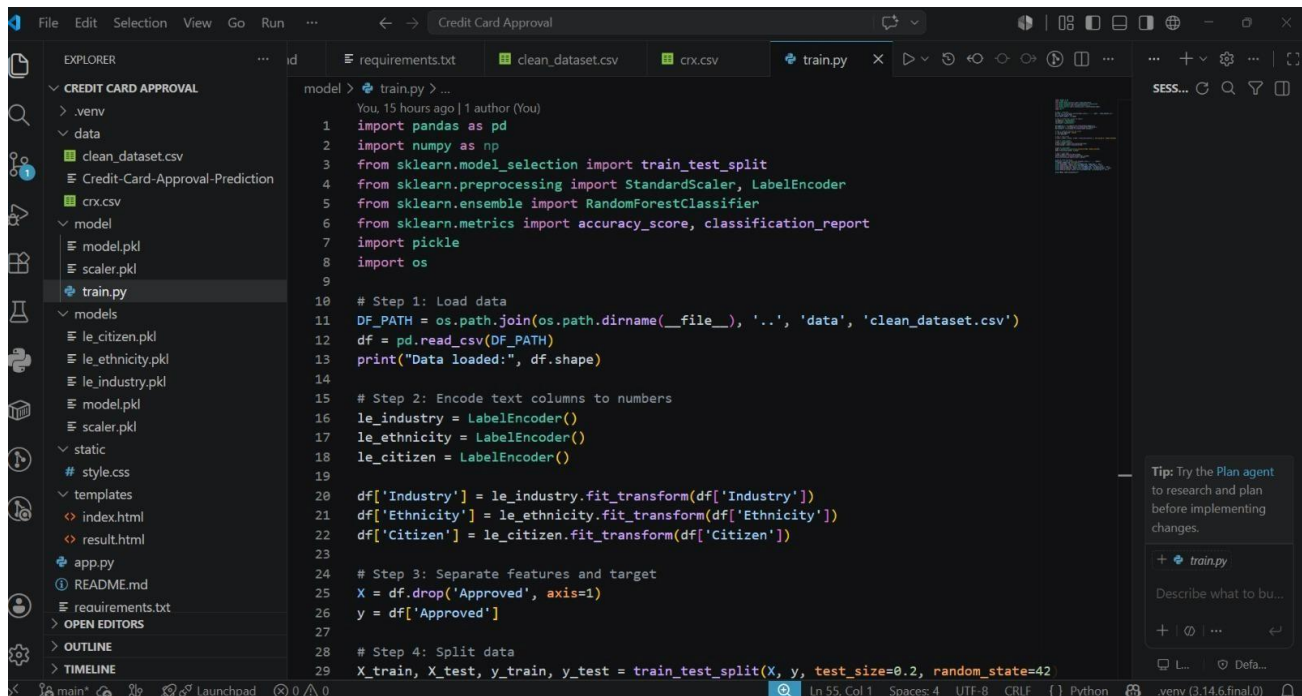
```
python app.py
```

5. Open the URL displayed in the terminal (usually <http://127.0.0.1:5000>) in your web browser.
6. Enter the required applicant details, such as age, income, employment status, credit score, and other information.
7. Click the **"Predict Approval"** button.
8. The system will process the input using the trained **Random Forest** machine learning model and display the prediction as **Approved** or **Rejected**, along with the confidence score.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Prediction accuracy depends on the quality and size of the training dataset.	Use a larger, more diverse, and better-quality dataset to improve accuracy.
2	The application currently runs only on the local system using the Flask development server.	Deploy the application on Render, PythonAnywhere, or another cloud platform.
3	Applicant details are not stored in a database.	Integrate a database such as MySQL or SQLite in future versions.
4	The system does not provide user login or authentication.	Add user authentication and authorization in future updates.
5	The model currently uses only the Random Forest algorithm.	Compare and optimize performance using algorithms such as XGBoost, Gradient Boosting, or Neural Networks.

Code Implementation Screenshots

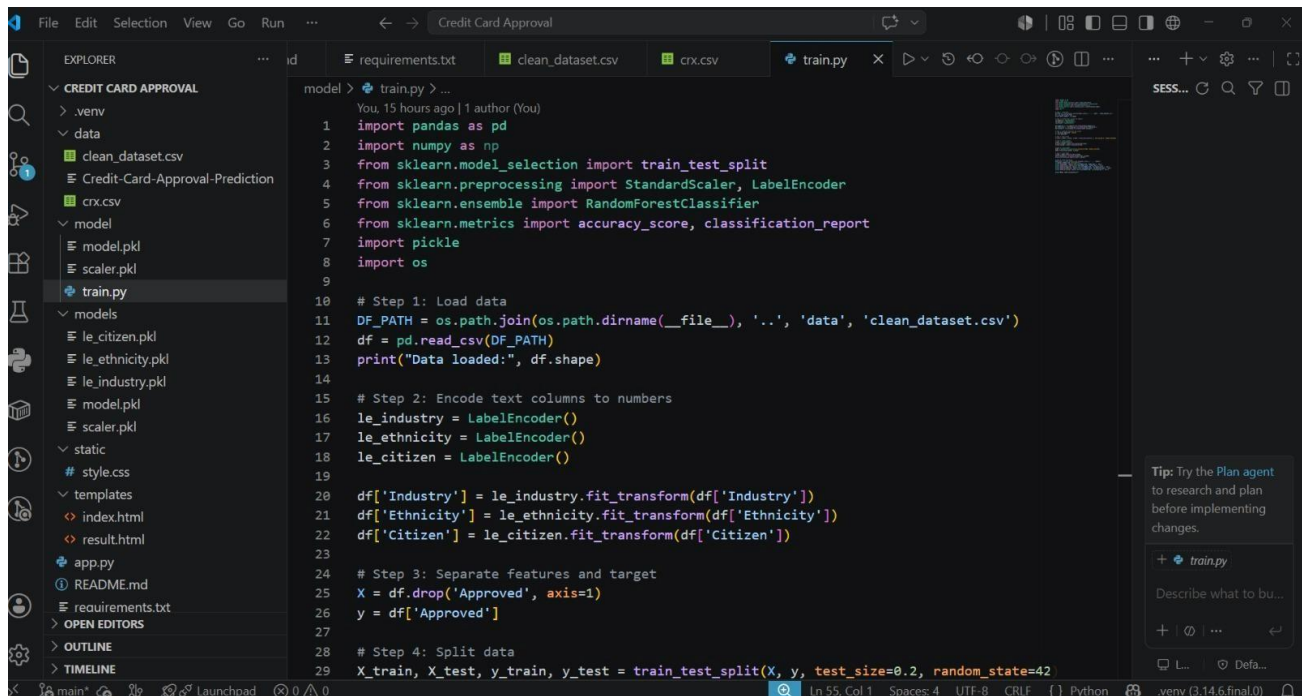


The screenshot shows a VS Code editor window with the file explorer on the left, the main editor in the center, and a sidebar on the right. The file explorer shows a project named 'CREDIT CARD APPROVAL' with a 'data' folder containing 'clean_dataset.csv' and 'crx.csv', and a 'model' folder containing 'model.pkl', 'scaler.pkl', and 'train.py'. The main editor shows the 'train.py' file with the following code:

```

1  import pandas as pd
2  import numpy as np
3  from sklearn.model_selection import train_test_split
4  from sklearn.preprocessing import StandardScaler, LabelEncoder
5  from sklearn.ensemble import RandomForestClassifier
6  from sklearn.metrics import accuracy_score, classification_report
7  import pickle
8  import os
9
10 # Step 1: Load data
11 DF_PATH = os.path.join(os.path.dirname(__file__), '..', 'data', 'clean_dataset.csv')
12 df = pd.read_csv(DF_PATH)
13 print("Data loaded:", df.shape)
14
15 # Step 2: Encode text columns to numbers
16 le_industry = LabelEncoder()
17 le_ethnicity = LabelEncoder()
18 le_citizen = LabelEncoder()
19
20 df['Industry'] = le_industry.fit_transform(df['Industry'])
21 df['Ethnicity'] = le_ethnicity.fit_transform(df['Ethnicity'])
22 df['Citizen'] = le_citizen.fit_transform(df['Citizen'])
23
24 # Step 3: Separate features and target
25 X = df.drop('Approved', axis=1)
26 y = df['Approved']
27
28 # Step 4: Split data
29 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
  
```

The sidebar on the right shows a 'Tip: Try the Plan agent to research and plan before implementing changes.' and a 'train.py' button.



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File Edit Selection View Go Run ... Credit Card Approval

EXPLORER

- CREDIT CARD APPROVAL
 - .venv
 - data
 - clean_dataset.csv
 - Credit-Card-Approval-Prediction
 - crx.csv
 - model
 - model.pkl
 - scaler.pkl
 - train.py
 - models
 - le_citizen.pkl
 - le_ethnicity.pkl
 - le_industry.pkl
 - model.pkl
 - scaler.pkl
 - static
 - style.css
 - templates
 - index.html
 - result.html
 - app.py
 - README.md
 - requirements.txt
 - OUTLINE
 - TIMELINE

model > train.py > ...

```

40 # Step 7: Test model
41 y_pred = model.predict(X_test_scaled)
42 print("Accuracy:", accuracy_score(y_test, y_pred))
43 print(classification_report(y_test, y_pred))
44
45 # Step 8: Save everything
46 MODEL_DIR = os.path.join(os.path.dirname(__file__), '..', 'models')
47 os.makedirs(MODEL_DIR, exist_ok=True)
48 pickle.dump(model, open(os.path.join(MODEL_DIR, 'model.pkl'), 'wb'))
49 pickle.dump(scaler, open(os.path.join(MODEL_DIR, 'scaler.pkl'), 'wb'))
50 pickle.dump(le_industry, open(os.path.join(MODEL_DIR, 'le_industry.pkl'), 'wb'))
51 pickle.dump(le_ethnicity, open(os.path.join(MODEL_DIR, 'le_ethnicity.pkl'), 'wb'))
52 pickle.dump(le_citizen, open(os.path.join(MODEL_DIR, 'le_citizen.pkl'), 'wb'))
53
54 print("Model saved successfully!")
55

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```

(.venv) PS D:\Credit Card Approval> & "d:\Credit Card Approval\.venv\Scripts\python.exe" "d:\Credit Card Ap
proval\model\train.py"

```

	precision	recall	f1-score	support
0	0.82	0.90	0.86	68
1	0.89	0.81	0.85	70
accuracy			0.86	138
macro avg	0.86	0.86	0.85	138
weighted avg	0.86	0.86	0.85	138

Model saved successfully!

(.venv) PS D:\Credit Card Approval>

Ln 55, Col 1 Spaces: 4 UTF-8 CRLF Python .venv (3.14.6.final.0)

Tip: Try the Plan agent to research and plan before implementing changes.

+ train.py

Describe what to bu...

File Edit Selection View Go Run ... Credit Card Approval

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app.py > predict

```

1 You, 15 hours ago | 1 author (You)
2 from flask import Flask, render_template, request
3 import pandas as pd
4 import pickle
5 import numpy as np
6
7 app = Flask(__name__)
8
9 # Load saved model and tools
10 model = pickle.load(open('models/model.pkl', 'rb'))
11 scaler = pickle.load(open('models/scaler.pkl', 'rb'))
12 le_industry = pickle.load(open('models/le_industry.pkl', 'rb'))
13 le_ethnicity = pickle.load(open('models/le_ethnicity.pkl', 'rb'))
14 le_citizen = pickle.load(open('models/le_citizen.pkl', 'rb'))
15
16 @app.route('/')
17 def home():
18     return render_template('index.html')
19
20 @app.route('/predict', methods=['POST'])
21 def predict():
22     # Get form data
23     gender = int(request.form['gender'])
24     age = float(request.form['age'])
25     debt = float(request.form['debt'])
26     married = int(request.form['married'])
27     bank_customer = int(request.form['bank_customer'])
28     industry = request.form['industry']
29     ethnicity = request.form['ethnicity']
30     years_employed = float(request.form['years_employed'])

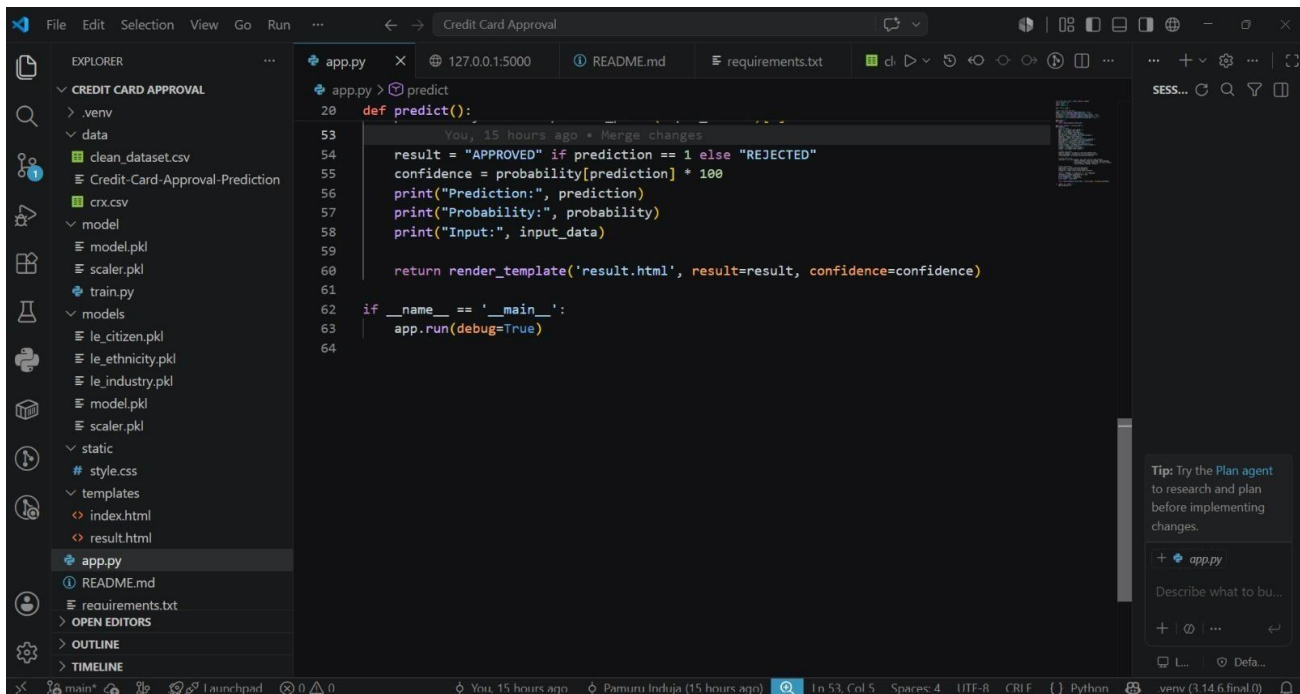
```

Ln 53, Col 5 Spaces: 4 UTF-8 CRLF Python .venv (3.14.6.final.0)

Tip: Try the Plan agent to research and plan before implementing changes.

+ app.py

Describe what to bu...



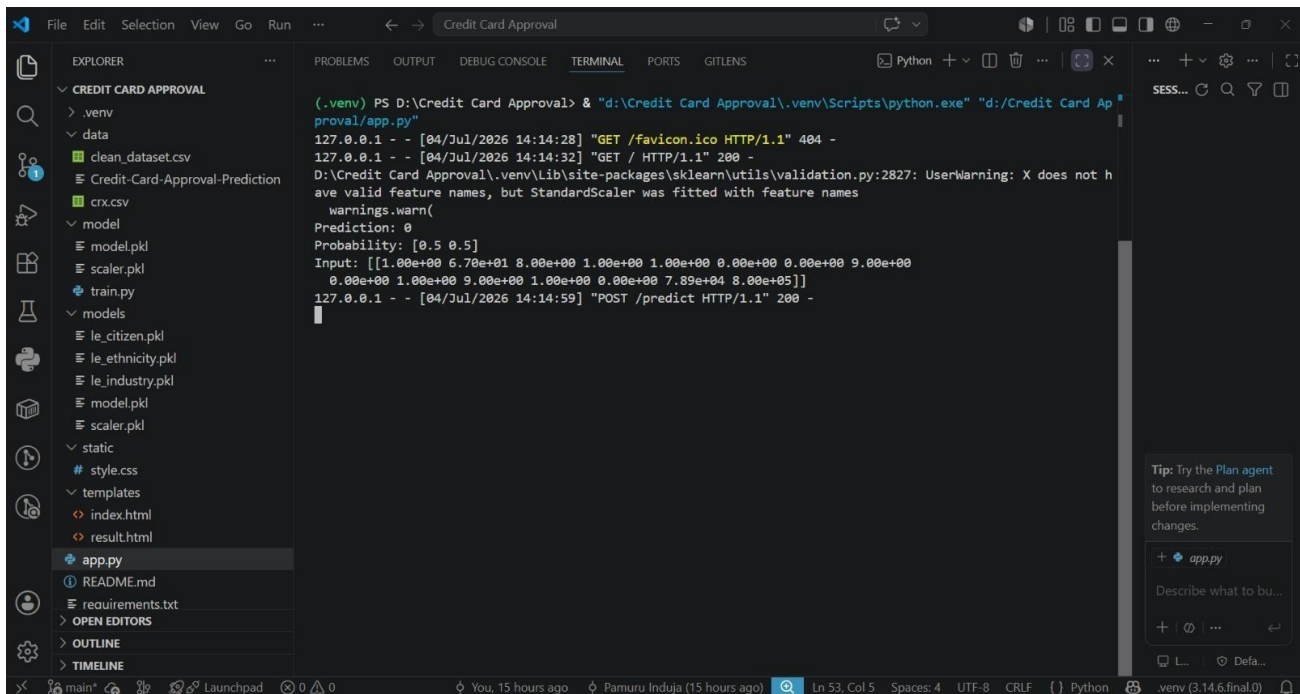
The image shows a VS Code editor window with the file explorer on the left and the editor area in the center. The file explorer shows a project named 'CREDIT CARD APPROVAL' with various files and folders. The editor area shows the 'app.py' file with the following code:

```

20 def predict():
21     You, 15 hours ago • Merge changes
22     result = "APPROVED" if prediction == 1 else "REJECTED"
23     confidence = probability[prediction] * 100
24     print("Prediction:", prediction)
25     print("Probability:", probability)
26     print("Input:", input_data)
27
28     return render_template('result.html', result=result, confidence=confidence)
29
30 if __name__ == '__main__':
31     app.run(debug=True)
32

```

The status bar at the bottom shows the file is 'app.py' in the 'CREDIT CARD APPROVAL' project, with a Python interpreter selected. The right sidebar shows a 'SESS...' panel with a tip to try the Plan agent.



The image shows the same VS Code editor window, but with the 'TERMINAL' panel open at the bottom. The terminal shows the output of the application running. The output includes a warning about feature names and the prediction results:

```

(.venv) PS D:\Credit Card Approval> & "d:\Credit Card Approval\.venv\Scripts\python.exe" "d:\Credit Card Ap
proval/app.py"
127.0.0.1 - - [04/Jul/2026 14:14:28] "GET /favicon.ico HTTP/1.1" 404 -
127.0.0.1 - - [04/Jul/2026 14:14:32] "GET / HTTP/1.1" 200 -
D:\Credit Card Approval\.venv\Lib\site-packages\sklearn\utils\validation.py:2827: UserWarning: X does not h
ave valid feature names, but StandardScaler was fitted with feature names
  warnings.warn(
Prediction: 0
Probability: [0.5 0.5]
Input: [[1.00e+00 6.70e+01 8.00e+00 1.00e+00 1.00e+00 0.00e+00 0.00e+00 9.00e+00
0.00e+00 1.00e+00 9.00e+00 1.00e+00 0.00e+00 7.89e+04 8.00e+05]]
127.0.0.1 - - [04/Jul/2026 14:14:59] "POST /predict HTTP/1.1" 200 -

```

The status bar at the bottom shows the file is 'app.py' in the 'CREDIT CARD APPROVAL' project, with a Python interpreter selected. The right sidebar shows a 'SESS...' panel with a tip to try the Plan agent.